CASE STUDY

# ONCORE & proprioception

## A case study in neurodiverse swim instruction

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| LEARNER<br>Brae, 6 years old | DATE PREPARED<br>25 July 2026 · updated 4 October 2026 |
| PREPARED BY<br>Connor Bailey — Assistant Manager, Darren Lange Swimming Academy | INTENDED USE<br>Teacher professional development |



Brae with Connor and ONCORE at Darren Lange Swimming Academy.

## Executive summary

Brae is a 6-year-old swimmer with ASD and ADHD. At the start of this work he could not breathe bilaterally in freestyle on his own. To get air he dropped into a backstroke-style kick, and the rest of the stroke went with it: his arms barely cleared the water, the kick stopped, and his head craned forward instead of rotating.

Connor introduced ONCORE in Brae's private lessons on 13 July 2026. Eight sessions followed on the same drill progression. Rotation was the limiter throughout, so the drills concentrated on kicking through the rotation and on alternating arms, and every lesson finished with a swim to see what had carried over.

The distance Brae could hold with acceptable technique went from 5 metres in the first session to 10 metres in the second. On 7 September he swam 12.5 metres of bilateral freestyle, with proficient head position and correct timing between his arms and his kick. On 14 September he swam a session with no ONCORE at all.

OUTCOME AND IMPACT

Across eight sessions Brae went from 5 metres of workable freestyle to 12.5 metres, and then to a session without ONCORE. The same three focuses ran every week — bubbles, kicking, three arms — on the same drills, in the same order.

This pattern of repetition has benefitted Brae immensely. We have slowly chipped away at what seemed an insurmountable task, and given him the confidence to swim well.

## Learner background

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| **Age** | 6 years old |
| **Diagnosis** | ASD and ADHD |
| **Lesson format** | 1:1 private lessons. |
| **Time with DLSA and this teacher** | Taught by Connor since Brae was 3, across 7 levels of the DLSA program, from Advanced Toddlers to LTS4. |

## The challenge

Learning to swim carries more weight for a child like Brae than for most. Guan and Li (2017) examined 23 unintentional drowning deaths of children with ASD under 15. Wandering preceded 73.9% of them.

Brae's main difficulty was bilateral breathing in freestyle. Without a proprioceptive cue for where his arms and body were, he could not coordinate the rotation a side breath needs. He took the air by dropping into a backstroke-style kick instead.

The rest of the stroke followed. His arms barely cleared the surface, the kick stopped almost entirely, and his head tilted forward rather than rotating.

> **“One day he said he didn't know where his arms were.”**
>
> BRAE · AS RECALLED BY CONNOR

That is why Connor reached for ONCORE. Brae had named the problem himself: he could not feel where his arms were in the water.

## The approach: ONCORE

**Disclosure:** Since hearing of Connor's findings and work in this space, Jade Edmistone, Managing Director of Get Oncore Pty Ltd, gifted the ONCORE used in these lessons and offered an affiliate link for their products. The link was not active at the time of printing. Connor is not paid for this case study.

ONCORE® is a skill acquisition tool made from low-density foam with a textured surface. The ONCORE Academy says the texture stimulates tactile receptors and gives swimmers immediate feedback on arm position, core engagement and body alignment in the water.

They market ONCORE as supporting proprioception — the body's sense of where it is and how it is moving — and “water feel” for swimmers of all levels. No independent published research on ONCORE itself could be located, so treat any claim that it is proven as unverified. (uncertain)

Connor introduced ONCORE to give Brae something to feel rather than something to be told. Verbal and visual correction had not shifted the breathing pattern, so ONCORE replaced it with direct tactile feedback on where his arms and body were in the water.

## Implementation

**SESSION 1** **13 July 2026 · private lesson**

### Drill progression

- Freestyle kick with arms extended in front — 15 minutes.
- Alternating arms, front and back — 10 minutes.
- Freestyle swim — 5 minutes.

### Observations

- Body position improved, though his head still tilted forward.
- Arm pattern became predictable.
- Breath position improved — not full backstroke, but not fully conventional either.

- Held reasonable technique for the first 5 metres. Over the last 7 metres he did not revert to backstroke, but did start lifting his head rather than rolling to breathe.

**SESSION 2** **20 July 2026 · private lesson**

Same drill progression, self-paced. Brae told Connor when he was ready to attempt his arms without ONCORE.

### Observations

- Further improvement to body position and kicking ability.
- Arm pattern improved further, and his breath was consistent over 10 metres.
- In the last 2 metres his breath position was wonky, but noticeably better than his baseline pattern.

**SESSIONS 3–6** **27 July to 31 August 2026 · private lessons**

No lessons on 3 or 17 August. Connor was on leave.

The same drill progression ran across these four sessions. Rotation stayed the limiter: Brae was not rotating far enough, so the drills concentrated on kicking through the rotation and on alternating arms across 12.5 metres. Each lesson finished with a swim to see how much had carried over.

**SESSION 7** **7 September 2026 · private lesson**

### Observations

- Started on drills with ONCORE, then moved to swimming.
- Swam 12.5 metres of bilateral freestyle.
- Proficient head position both when blowing bubbles and when breathing, with correct timing between his arms and his kick.

**SESSION 8** **14 September 2026 · private lesson**

### Observations

- First session with no ONCORE at all.
- 12.5 metres, his normal distance.
- Kicking was better than his original pattern.
- He did not blow bubbles at any point in the session.

- He arrived distracted and his listening was not at its usual level. The lesson finished early.

## Results

Brae's distance with acceptable technique went from 5 metres on 13 July to 10 metres on 20 July, and to 12.5 metres of bilateral freestyle on 7 September, with proficient head position and correct timing between his arms and his kick. On 14 September he covered the same distance with no ONCORE at all, and his kick was better than his original pattern, though he did not blow bubbles at any point that session.

The breathing pattern holds with ONCORE. Whether it holds without it is still open after one session without it.

## Discussion

Brae's time on task, and his ability to process, focus and perform, was a major factor in moving him to private lessons. He was unsafe in a group environment. I was extremely strict with him, on a three-warning-then-out system. That structure and discipline helped, but I was still at an impasse with his breathing.

One day he said he didn't know where his arms were, so I thought of ONCORE. It was vital to how I taught Brae. He built awareness of where his body was in the water, and his distance went up quickly and easily.

I feel that had I not used ONCORE with Brae, his frustration at not understanding what we needed to do would have made teaching him much harder.

## What the research says

Research on proprioception, tactile feedback, motor learning and swimming instruction for autistic children helps explain why this approach suited Brae. It cannot prove what caused his result. Each row links a moment from the lessons to the research behind it.

| WHAT HAPPENED WITH BRAE | WHAT THE RESEARCH SAYS |
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### Telling him was not working

Verbal and visual correction had not shifted the breathing pattern before ONCORE came out.

Shafer et al. (2021) compared 40 autistic participants aged 10 to 33 with 25 matched controls on a precision grip task. The autistic participants relied more on visual feedback and less on proprioceptive feedback, which is consistent with reduced integration of proprioceptive input.

That study tested fine motor control in a laboratory, not swimming, and it did not test this tool. It gives a reason to train the proprioceptive channel directly rather than proof that doing so works in a pool.

### He could not feel where his arms were

Brae said so himself, and that is what prompted Connor to try ONCORE.

Kemp et al. (2024b) tested sensory processing against water competency in 37 autistic children aged 5 to 9. Children who scored higher for low registration — missing sensory input that is there — scored lower on both water competency assessments ($\rho = -.33$).

The same study found that children with more exposure to group swim lessons scored higher on water competency, so it does not support private lessons over group teaching. It is a correlation across 37 children, not a finding about any one of them.

### Moving to private lessons

Brae was unsafe in a group environment, so lessons moved to a 1:1 format.

Kemp et al. (2024a) ran a randomised controlled trial with 37 autistic children aged 5 to 9: ten weekly sessions of 60 minutes, one instructor per child. Water competency improved significantly, with effect sizes between 0.63 and 0.87.

The trial did not compare 1:1 against group teaching, so it shows the format works rather than that it works better.

### Something to feel instead

ONCORE gave Brae tactile feedback on where his arms and body were in the water.

Cuppone et al. (2016) added vibro-tactile feedback to robot-assisted proprioceptive training. Somatosensory accuracy and motor performance improved more than with proprioceptive training alone, and the gains carried across to movements that had not been trained directly.

This was a robot-assisted arm task with adults, not a swimming drill.

### Rotation was the limiter

Drills concentrated on kicking through the rotation and on alternating arms, week after week.

Khiyami et al. (2022) ran a six-week randomised controlled trial with 18 young male swimmers. The core-training group improved their 50 m freestyle time by 1.4 seconds on average, along with swimming velocity, stroke length, stroke index and the contraction timing of key trunk muscles.

That supports the training principle ONCORE works on. It says nothing about ONCORE itself.

### Coordination as well as autism

Brae has ADHD alongside ASD, and coordination was the presenting problem.

Meachon et al. (2025) surveyed the parents of 168 children with ADHD aged 8 to 17. 81.5% scored in the probable range for developmental coordination disorder on a parent questionnaire, and 27.3% already had a DCD diagnosis.

That is parent report rather than assessment, in an older age group than Brae, and Brae has no DCD diagnosis. It applies by analogy only.

### The same drills, every week

One drill progression and three focuses — bubbles, kicking, three arms — across eight sessions.

Muharib and Ramirez (2026) reviewed 17 experimental studies teaching swimming and water safety to autistic people, covering 107 participants. Behavioural skills training — instruction, modelling, rehearsal and feedback — featured in 82% of the interventions, and the largest effects came from combining it with systematic prompting (mean Tau-U 0.83).

Most of the included studies were single-case designs, and the review is an advance online publication. (uncertain)

### Taking ONCORE away

On 14 September Brae swam his normal 12.5 metres without ONCORE, but did not blow bubbles at any point.

The transfer finding in Cuppone et al. (2016) is the reason to test without the tool rather than assume the skill has carried.

One session is not enough to tell whether Brae's pattern has transferred, and the bubbles suggest part of it has not.

LIMITS OF THIS EVIDENCE

Brae is one learner across eight sessions, with no comparison group and no control. Other things changed over those nine weeks: he is 6 and still developing, the same drills repeated week on week, and Connor has taught him since he was 3.

None of the research above tested ONCORE. The swimming studies show that structured, repeated teaching works with autistic children, and the sensory research explains why a tactile approach might suit a swimmer with Brae's profile. Neither shows that ONCORE caused what happened here. The test is whether the pattern holds across further sessions without it.

## Practical takeaways for teachers

### Swap telling for feeling

When verbal and visual correction has stopped working, a tactile cue gives the swimmer something to feel instead of something to remember. That is why ONCORE came out for Brae.

### Hold the same focuses every week

Brae had three: bubbles, kicking, three arms. Same drills, same order, eight sessions. His progress came alongside that consistency rather than out of any single lesson.

### Finish with a swim

Drills show you what a swimmer can do with the tool in their hands. The swim at the end shows you what carried over. Every one of Brae's lessons finished that way.

### Take the tool away and check

A skill that only turns up with ONCORE has not transferred yet. Brae held his distance without it and lost his bubbles, which told us more than another session with them would have.

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# Consent & media notes

Media included: one photo of Brae and Connor with ONCORE.

Written parental consent is on file for use of Brae's first name and the photograph in this case study and the accompanying slides, per DLSA case study conventions.

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